

$$\textcircled{1} \quad \text{Maybe}(\alpha) \equiv \forall \beta. (\alpha \rightarrow \beta) \rightarrow \beta \rightarrow \beta$$

cf. Haskell:

$$f :: \text{Maybe Int} \rightarrow \text{String}$$

$$f (\text{Just } n) = \text{show } n$$

$$f \text{ Nothing} = \text{"error"}$$

$$\alpha = \text{Int}$$

$$\beta = \text{String}$$

$$\begin{array}{c} \text{show: Int} \rightarrow \text{String} \\ \alpha \rightarrow \beta \end{array}$$

$$\begin{array}{c} \text{"error"} : \text{String} \\ \beta \end{array}$$

$$g: \alpha \rightarrow \overset{\text{IN}}{\beta}, \quad N: \overset{\text{IN}}{\beta}, \quad m: \text{Maybe}(\alpha)$$

$$\vdash m\{\text{IN}\} g N : \text{IN}$$

② ~~Ata~~ Perhaps I have $t : \alpha$

$$\text{Just}_\alpha t = \lambda \beta. \lambda f : (\alpha \rightarrow \beta). \lambda n : \beta. f t$$

Perhaps I don't have any α 's

$$\text{Nothing}_\alpha = \lambda \beta. \lambda f : (\alpha \rightarrow \beta). \lambda n : \beta. n$$

$$(\lambda x. x) (\text{Just}_\alpha t)$$

data Fruit = Apple | Pear | Banana
 T T F

$$\text{Fruit} = \forall \beta. \beta \rightarrow \beta \rightarrow \beta \rightarrow \beta$$

$$\text{isGreen} : \text{Fruit} \rightarrow \mathbb{B}$$

$$\text{isGreen} = (\lambda x : \text{Fruit}. x \{ \mathbb{B} \} \text{ true } \text{ true } \text{ false})$$

$$\text{Apple} \equiv \lambda \beta. \lambda x_1 : \beta. \lambda x_2 : \beta. \lambda x_3 : \beta. x_1$$

⑤

data IntList = Nil | Cons Int IntList

length :: IntList $\rightarrow$ Int

length Nil = 0

length (Cons n ns) = 1 + length ns

$\mathbb{N} \text{ NatList} = \forall \beta. \beta \rightarrow (\mathbb{N} \rightarrow \overset{\beta}{\text{NatList}} \rightarrow \beta) \rightarrow \beta$

length = $\lambda l : \text{NatList}. l \{ \mathbb{N} \} \text{ zero } (\lambda n : \mathbb{N}. \lambda s : \mathbb{N}. \text{succ } s)$

$$(4) \quad \text{xor} = \lambda x. \lambda y. x (\neg y) y$$

$$\text{where } \neg = \lambda b. b \text{ false true}$$

$$\text{false} = \lambda x. \lambda y. y$$

$$\text{true} = \lambda x. \lambda y. x$$

Whenever $b_1, b_2 \in \{\text{false}, \text{true}\}$

$$\text{xor } b_1 \ b_2 \longrightarrow_v^* [b_1 \text{ XOR } b_2]$$

T	T	F
T	F	T
F	T	T
F	F	F

{Without xor true false $\rightarrow_v$ etc?}

Let $b_1, b_2 \in \{\text{true}, \text{false}\}$

~~xxx~~

$$(\text{xor } b_1) \ b_2 \quad (\bullet \ b_2)$$

$$\longrightarrow_v (\lambda y. b_1 (\neg y) y) \ b_2 \quad (\bullet)$$

$$\longrightarrow_v (b_1 (\neg b_2)) \ b_2 \quad ((b_1 \bullet) \ b_2)$$

$$\longrightarrow_v (b_1 ((b_2 \text{ false}) \text{ true})) \ b_2 \quad ((b_1 (\bullet \text{ true})) \ b_2)$$

⑤ Case-split on what b_2 is to continue.

Case 1: $b_2 = \text{true}$

~~$b_1 (b_2 \text{ true})$~~

$= b_1 (\text{true false true}) \text{ true}$

$\rightarrow \dots$

Case 2: $b_2 = \text{false}$.

~~$b_1 \text{ true} = b_2$~~

~~$\text{true} (\text{true false } \vdots)$~~

$\neg \text{true} \rightarrow_v^* \text{false}$

1st

$(\lambda b. b \text{ false true}) \text{ true} \rightarrow_v \text{true false true} \quad \beta$

⑥

$$\text{true} = \lambda x. \lambda y. x$$

(• ~~False~~^{true})

$$\begin{array}{c} 2^{\text{nd}} \quad \text{true false} \rightarrow_v \lambda y. \text{false} \\ \hline (\text{true false}) \text{ true} \rightarrow_v (\lambda y. \text{false}) \text{ true} \quad \text{APP}_1 \end{array}$$

$$\begin{array}{c} 3^{\text{rd}} \\ \hline (\lambda y. \text{false}) \text{ true} \rightarrow_v \text{false} \end{array}$$

$$\neg \text{true} \rightarrow_v^* \text{false}$$